

LKB-FOS Update 2026

John Carroll
University of Sussex, UK

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Outline

One LKB-FOS release since last year's summit, on 24 July

Bug fixes and enhancements

Improvements to scalability

Support for DELPH-IN unified grammar configuration file format

Issues and Enhancements

A few bug fixes and enhancements

- fixed error on starting `[incr tsdb()]` podium The value `:ALL` is not of type `(OR STRING CONS (MEMBER :WILD NIL))` when binding `:DIRECTORY`
- post-generation chart mapping is now applied to 'full' FS rather than 'edge' FS (can make a difference when information is percolated top-down)
- implemented full forest `[incr tsdb()]` output (experimental and code not yet integrated)

Queries answered on Discourse site about: `irregs.tab` problem, keyboard entry of non-Western-European character sets, case sensitive input

Continued to clean up code and improve comments

Scalability

Reduced memory usage for parsing and generation

- increased sharing of arcs between feature structures
- eliminated an $O(n)$ factor in chart edge size
- reduced ‘over-copying’ of FSES in token mapping

Improved memory behaviour of LKB process

- refactored code for discarding temporary data – now used in LKB batch parsing and generation, and [incr tsdb()] batch runs
- memory efficiently released back to operating system to control process size

On a 32GB iMac, 6 concurrent LKB-FOS processes computing full forests for ‘Cathedral and Bazaar’ with 1M-edge limit get 4.5x speed-up over a single process

DELPH-IN Unified Grammar Configuration File Format

Grammars may now be loaded using the TDL-inspired DELPH-IN unified configuration file format (e.g., for the ERG, via `config.tdl` and `english.tdl`)

Based on RFCs by Michael Goodman in <https://github.com/delph-in/docs/wiki/>

- `:begin / :end :type / :instance` and `:include` are implemented as in `TdlRFC`
- `:begin / :end :config` is implemented as in `GrammarConfigurationRfc`, and may be conditionalised following Sub-Proposal 1 "Another possible syntax ..."

Extends `TdlRFC` and `ACE` and `PET` implementations

- further `:instance` status values `label` and `root`
- further parameters beyond those in `AceConfiguration`, including: `load-lkb-preferences`, and `preload- / postload-lisp-files`

Tested on ERG, based on `ace/config.tdl` and `english.tdl`, merging into them `lkb/script` and parts of `lkb/globals.lsp`

;;; Files to load

lkb/config.tdl

```
grammar-top                := "../english-lkb.tdl".
variable-property-mapping := "../semi.vpm".
maxent-model               := "../redwoods.mem".

:begin :config ACE.
preprocessor               := "../rpp/tokenizer.rpp".
preprocessor-modules       := ../rpp/xml.rpp ../rpp/ascii.rpp ../rpp/lgt.rpp
  ../rpp/quotes.rpp ../rpp/wiki.rpp ../rpp/gml.rpp ../rpp/html.rpp. ; !!! invalid TDL
:end :config.
:begin :config LKB.
preprocessor               := "../rpp/setup.lsp".
:end :config.
...

:begin :config ACE.
post-model-path            := "english-postagger.hmm".
:end :config.

:begin :config LKB.
load-lkb-preferences       := "user-prefs.lsp".
preload-lisp-files        := "globals.lsp" "user-fns.lsp" "patches.lsp" "settings.lsp".
postload-lisp-files       := "mrsglobals.lsp".
:end :config.
```

```
;;; Parameters

;; Types
top-type          := *top*.
non-idiom-root    := root_non_idiom.

;; Feature Paths
orth-path         := ORTH.
label-path        := LNAME.

;; Lists
list-type         := *list*.
cons-type         := *cons*.
...

;; Processing
parsing-roots     := root_strict root_frag.
generation-roots  := root_gen_nofocus.
deleted-daughters :=
  ARGS HD-DTR NH-DTR LCONJ-DTR RCONJ-DTR DTR DTR1 DTR2 DTRA DTRB.
...

;; Token mapping types and paths
...

;; Chart mapping rule features
...
```

```
:begin :type.  
:include "fundamentals".  
...  
:include "mtr". ; !!! loaded here with rest of types, not from config.tdl  
:include "letypes". ; !!! in lkb/script these leaf types are cached  
:end :type.  
  
:begin :instance :status token-mapping-rule. ; !!! previously inside type environment  
:include "tmr/gml".  
:include "tmr/ptb".  
...  
:end :instance.  
...  
:begin :instance :status generic-lex-entry.  
:include "gle".  
:include "gle-gen". ; !!! not in lkb/script  
:end :instance.  
...  
:begin :instance :status label. ; !!! :status label added  
:include "parse-nodes". ; !!! duplicates parse-node-labels parameter in config.tdl  
:end :instance.  
  
:begin :instance :status root. ; !!! :status root added  
:include "roots".  
:end :instance.
```


Several practical issues

- LKB cannot load any other constructs before type hierarchy is finalised, so only `:include` statements are accepted inside `:begin / :end :type` environment
- similar restriction on lexical entry environment
- are conditionals too verbose? i.e., `:begin :config system. ... :end :config.`
- between LKB and ACE some important parameters have exact counterparts, but many more do not; even less overlap with PET
- no standard notation for parameter values that are paths or empty lists
- do repeated parameter settings cause conflict or reassignment?
- other oddities, including those noted !!! above

Suggest discussing these and further issues in a SIG later this week

Summary

Reasonable progress over the past year

- a few fixes and enhancements
- improved scalability
- support for DELPH-IN unified grammar configuration file format

Main items on to-do list

- move to new Wayland and SDL2 backends for GUI
- improve support for non-GUI use